

Basic Counting Principles

r-Permutation An r-permutation of A is a way of arranging any r of the objects of A in a row, with number denoted as $P_r^n = \frac{n!}{(n-r)!}$ by MP.

Principle of Division If we can partition instances of an event E into disjoint r-element groups which each gives rise to a unique E' , then $|E'| = \frac{|E|}{r}$.

Circular Permutation An r-circular permutation of A is a circular permutation of r distinct objects taken from A , denoted $Q_r^n = \frac{P_r^n}{r}$.

Combination An r-combination of A is an r-element subset of A , denoted by $C_r^n = \binom{n}{r} = \frac{P_r^n}{r!} = \frac{n!}{r!(n-r)!}$. Additionally, $\binom{n}{r} = \binom{n}{n-r}$.

Combinatorial Identities

- $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$ (Either take the current element, or choose r from the set without the current element)
- $\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}$ (Move r over, then LHS becomes choosing r then a leader while RHS becomes choosing the leader then remaining $r-1$)
- $\binom{n}{r} = \frac{n-r+1}{r} \binom{n}{r-1}$, $\binom{n}{r} = \frac{n}{n-r} \binom{n-1}{r}$, $\binom{n}{m} \binom{m}{r} = \binom{n}{m-r} \binom{m-r}{r}$

Principle of Inclusion and Exclusion $|A_1 \cup A_2 \cup \dots \cup A_q| = \sum_{i=1}^q |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| \dots + (-1)^{q-1} |A_1 \cap \dots \cap A_q|$

Euler's φ Function $\varphi(n)$ denotes the number of integers between 1 and n which are coprime to n , and $\varphi(n) = n(1 - \frac{1}{p_1}) \dots (1 - \frac{1}{p_k}) = n \prod_{i=1}^k (1 - \frac{1}{p_i})$.

Functions A to B : Injection: $|A| \leq |B|$, Surjection: $|A| \geq |B|$, Bijection: Equal
r-Combinations Without Consecutive $\binom{n-r+1}{r}$.

Twelfold Way

Objects / Boxes	Arbitrary	Injective	Surjective
Distinct / Distinct	n^r	$P(n, r) = \frac{n!}{(n-r)!}$	$n! S(r, n)$
Distinct / Identical	$\sum_{k=0}^n S(r, k)$	$\begin{cases} 1 & r \leq n \\ 0 & r > n \end{cases}$	$S(r, n)$
Identical / Distinct	$\binom{r+n-1}{r}$	$\binom{n}{r}$	$\binom{r-1}{n-1}$
Identical / Identical	$\sum_{k=0}^n p(r, k)$	$\begin{cases} 1 & r \leq n \\ 0 & r > n \end{cases}$	$p(r, n)$

- Injective / Surjective: Each box has at most / least one object.
- $S(r, n)$: Stirling numbers of the second kind.
- $p(r, n)$: Number of partitions of r identical objects into exactly n identical boxes.

Non-Negative Integer Solutions For the linear equation $x_1 + \dots + x_n = r$ where $x_i \geq 0$, the number of solutions is $H_r^n = \binom{n+r-1}{r}$ by stars and bars.

Multiset $M = \{r_1 \cdot a_1, r_2 \cdot a_2, \dots, r_n \cdot a_n\}$ where r_i is the repetition number of object a_i . $\infty \cdot a$ is used to indicate infinite copies of an object.

Multiset Permutations The number of r-permutations of $\{\infty \cdot a_1, \dots, \infty \cdot a_n\}$ is n^r . If $r = \sum_{i=1}^n r_i$, the number of permutations of M ,

$$P(r; r_1, \dots, r_n) = \frac{r!}{r_1! \dots r_n!}$$

Combinations with Repetition A m element multi-subset of M is a subset $\{m_1 \cdot a_1, \dots, m_n \cdot a_n\}$ where $\sum_{i=1}^n m_i = m$. The number of r-element multi-subsets is given by $\binom{r+n-1}{r}$.

Stirling Numbers of the Second Kind $S(r, n)$ or $\left\{ \begin{smallmatrix} r \\ n \end{smallmatrix} \right\}$ counts the number of ways to partition a set of r distinct objects into n **non-empty**, unlabeled subsets. Base Cases:

$$S(0, 0) = 1, S(r, 0) = 0, r > 0, S(0, n) = 0, n > 0, S(r, n) = 0, r < n.$$

Recursive Definition: $S(r, n) = nS(r-1, n) + S(r-1, n-1)$; either the r th element is in its own subset or it joins an existing subset.

Explicit Formula $S(r, n) = \frac{1}{n!} \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} j^r$. Derived from $n! \cdot S(r, n) = \sum_{j=0}^n (-1)^j \binom{n}{j} (n-j)^r$, where LHS is number of surjective functions from r elements to n elements, while RHS is the same by considering PIE on the set of functions avoiding element j . $S(n, k) \sim \frac{k^n}{k!}$ as $n \rightarrow \infty$

Partitions of an Integer A partition of r is writing r as a sum of positive integers where **order does not matter**. The number of partitions of r is $p(r)$, and $p(r, n)$

if into n sums. $p(n, 2) = \lfloor \frac{n}{2} \rfloor$, $p(n, 3) = \lfloor \frac{n^2+3}{12} \rfloor$

- $p(0, 0) = 1, p(r, 0) = 0, r > 0, p(0, n) = 0, n > 0$
- $p(r, n) = p(r-1, n-1) + p(r-n, n)$: Either one summand is 1, or all of them are at least 2 so we can subtract 1 from all.

Sequences to Polynomials $P(x) = a_0 + a_1x + \dots + a_nx^n$ represents the finite sequence a_n .

Binomial Coefficients $\binom{n}{r} = \frac{n!}{r!(n-r)!}$'s are known as the binomial coefficients.

Binomial Theorem $\forall n \geq 0, (x+y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$. Consider the string of length n with letters x or y .

Example Identities $\sum_{r=0}^n \binom{n}{r} = 2^n$ using $(1+1)^n$, $\sum_{r=0}^n (-1)^r \binom{n}{r} = 0$ using $(1+(-1))^n$, $\binom{n}{0} + \dots + \binom{n}{2k} = \binom{n}{1} + \dots + \binom{n}{2k+1} = 2^{n-1}$

Another Identity $\sum_{r=1}^n r \binom{n}{r} = n \cdot 2^{n-1}$ by differentiating $(1+x)^n$ and subbing in $x=1$

Multinomial Coefficients $\binom{n}{n_1, n_2, \dots, n_m} = \binom{n}{n_1} \binom{n-n_1}{n_2} \dots = \frac{n!}{n_1! n_2! \dots n_m!}$.

Multinomial Theorem For $n, m \in \mathbb{N}$, $(x_1 + x_2 + \dots + x_m)^n = \sum \binom{n}{n_1, n_2, \dots, n_m} x_1^{n_1} x_2^{n_2} \dots x_m^{n_m}$ where we consider all m -ary sequences of nonnegative integers summing to n .

Ordinary Generating Functions

Sequences to Power Series $F(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{n=0}^{\infty} a_nx^n$ is the genfunc for the infinite sequence a_n .

Cauchy Product of Series The product of two series $F(x)G(x)$ corresponds to the sequence $c = (c_0, c_1, c_2, \dots)$, $c_r = \sum_{k=0}^r a_k b_{r-k}$. For multiple series, the product corresponds to $c_r = \sum_{k_1+k_2+\dots+k_m=r} a_{k_1}^{(1)} a_{k_2}^{(2)} \dots a_{k_m}^{(m)}$.

Weighted Sums To count solutions of $x_1 + 2x_2 + \dots + mx_m = r$, we can use generating functions with weights in exponents. For x_i with coefficient i , associate $F^{(i)}(x) = \sum_{k \in A(i)} x^{ik}$. Then $F(x) = F^{(1)}(x) \dots F^{(m)}(x)$ and the coefficient c_r of x^r counts solutions.

Cumulative Sums Let $F(x) = \sum_{n=0}^{\infty} a_nx^n$. Then $(1+x+x^2+\dots) \cdot F(x) = (a_0) + (a_0+a_1)x + \dots$ encodes $c_r = a_0 + \dots + a_r$.

Shifting a Sequence $x^m F(x) = \sum_{r=0}^{\infty} c_r x^{r+m}$ is the genfunc for the sequence where $c_r = a_{r-m}$.

Constrained Equation

$x_1 \geq 1 \Rightarrow F_1(x) = x + x^2 + x^3 + \dots$, $x_2 \leq 2 \Rightarrow F_2(x) = 1 + x + x^2$. The genfunc is $F(x) = F_1(x)F_2(x) \dots$. Then we can use coefficients to find the number of solutions. For $(1+x+\dots)^2$, the coefficient of x^k is $k+1$.

Dividing Genfuncs Use partial fraction decomposition instead, e.g. $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, $\frac{1}{1-2x} = \sum_{n=0}^{\infty} 2^n x^n$.

Partition Genfunc The genfunc for $p(n)$ is $\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} (1+x^k+x^{2k}+\dots) = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.

Euler's Partition Theorem Again $D(x) = \prod_{k=1}^{\infty} (1+x^k)$,

$O(x) = \prod_{k=1}^{\infty} \frac{1}{1-x^{2k-1}}$. Using $1+x^k = \frac{1-x^{2k}}{1-x^k}$, we get what we want.

Exponential Generating Functions The EGF for a sequence (a_r) is $a_0 + a_1 \frac{x}{1!} + \dots + a_r \frac{x^r}{r!} + \dots = \sum_{r=0}^{\infty} a_r \frac{x^r}{r!}$. $a_r = r!c_r$ where c_r is the r th coefficient of the genfunc.

OGF v/s EGF OGF are applicable in distribution or arrangement problems where ordering of objects involved is immaterial. EGF are useful in counting arrangements of objects where ordering is taken into account.

General EGF Formula Let a_r denote the number of r-permutations of the multi-set $\{n_1 \cdot b_1, \dots, n_k \cdot b_k\}$. Then the EGF is

$$(1 + \frac{x}{1!} + \dots + \frac{x^{n_1} b_1}{n_1!}) \dots (1 + \frac{x}{1!} + \dots + \frac{x^{n_k} b_k}{n_k!}).$$

Recurrence Relations

Linear Homogeneous Recurrence Relations RR with the form

$$c_0 a_n + c_1 a_{n-1} + \dots + c_r a_{n-r} = 0. \text{ A function } a_n = f(n) \text{ solves the RR.}$$

Characteristic Equation For a LHRR, replace a_i by x^i , then divide by x^{n-r} to get the characteristic equation. Any root of this equation is a characteristic root.

LHRR Theorem If $\alpha_1, \dots, \alpha_r$ are the distinct characteristic roots of our RR, then $a_n = A_1(\alpha_1)^n + \dots + A_r(\alpha_r)^n$ where A_i is constant is the general form of solutions to our RR. Sub in initial conditions to get the solution.

Complex Roots Express using $e^{ix} = \cos x + i \sin x$.

LHRR Theorem Again If $\alpha_1, \dots, \alpha_k$ are the distinct characteristic roots such that α_i has multiplicity m_i , the general form of the solution of our recurrence relation is $a_n = (A_{11} + A_{12}n + \dots + A_{1m_1} n^{m_1-1})(\alpha_1)^n + \dots + (A_{k1} + A_{k2}n + \dots + A_{km_k} n^{m_k-1})(\alpha_k)^n$.

General Linear Recurrence Relation A recurrence relation of the form

$$c_0 a_n + c_1 a_{n-1} + \dots + c_r a_{n-r} = f(n) \text{ is an rth order linear recurrence relation. An LHRR is a linear recurrence relation where } f(n) = 0.$$

If $g_1(n), g_2(n)$ are solutions for above, then $h(n) = g_1(n) - g_2(n)$ is a solution of the corresponding LHRR.

Solving General Linear Recurrence Relations

- Find the general solution $h(n)$ of the corresponding LHRR.
- Find a particular solution $g_0(n)$ of the linear recurrence relation. It can be chosen as a function of a similar type.
- The general family of solutions of $g_0(n)$ of the original recurrence relation is given by $g(n) = g_0(n) + h(n)$.
- Find the specific solution that fits the given initial conditions.

$g_0(n)$ Chart

- Ak^n , k is not characteristic root: Bk^n
- Ak^n , k is characteristic root with multiplicity m : $Bn^m k^n$
- $\sum_{i=0}^t p_i n^i$, 1 is not characteristic root: $\sum_{i=0}^t q_i n^i$
- $\sum_{i=0}^t p_i n^i$, 1 is a characteristic root of multiplicity m : $n^m \sum_{i=0}^t q_i n^i$
- $An^t k^n$, k is not a characteristic root: $(\sum_{i=0}^t q_i n^i) k^n$
- $An^t k^n$, k is a characteristic root of multiplicity m : $n^m (\sum_{i=0}^t q_i n^i) k^n$

Solving Linear Recurrence Relations with GF

- Transform equation into linear equation for generating function
- Solve linear equation and simplify the answer, often involving partial fractions
- $A(x) - x = \sum a_n x^n = \sum (a_{n-1} + a_{n-2}) x^n$, relate the genfunc to the recurrence and express the RHS in terms of $A(x)$, then make it the subject of the formula

Partial Fractions

- Distinct Linear Factors: $\frac{P(x)}{(1-r_1x)(1-r_2x)\dots(1-r_kx)} = \sum_{i=1}^k \frac{A_i}{1-r_ix}$
- Each term expands as $\frac{1}{1-rx} = \sum_{n=0}^{\infty} r^n x^n$
- Repeated Roots: Multiplicity m : $\frac{P(x)}{(1-rx)^m} = \frac{B_1}{1-rx} + \dots + \frac{B_m}{(1-rx)^m}$
- Expansion gives terms of the form $\frac{1}{(1-rx)^j} = \sum_{n=0}^{\infty} \binom{n+j-1}{j-1} r^n x^n$
- Use $x = \frac{1}{r_i}$ to eliminate and solve for other coefficients

System of Linear Recurrence Relations

- Domino Tiling: Tile $2 \times n$ board using 1×2 and 1×1 tiles
- Let a_n be complete tilings, and b_n be tilings with a missing square
- $a_n = b_n + a_{n-1} + b_{n-1} + a_{n-2}$
- $b_n = a_{n-1} + b_{n-1}$
- Use substitution to eliminate one sequence, then spam genfunc

Tutorial Stuff

More Identities

- $P_r^n = nP_{r-1}^{n-1}, P_r^n = (n-r+1)P_{r-1}^n$
- $P_r^n = \frac{n}{n-r}P_r^{n-1}, P_r^{n+1} = P_r^n + rP_{r-1}^n$
- $P_r^{n+1} = r! + r(P_{r-1}^n + P_{r-1}^{n-1} + \dots + P_{r-1}^r)$: Either take the last r elements, or place $n+1$, don't take $n+1, n, n-1, \dots, i+1$, then take i

Divisibility Argument

- Prove $(n+1)(n+2)\dots(2n)$ is divisible by 2^n
- Pairing $2n$ people tells us $\frac{(2n)!}{n! \cdot 2^n}$ is an integer. So $\frac{(2n)!}{n! \cdot 2^n} \cdot 2^n$ which is what we want is also an integer.

Even Function For any function $f(k), \sum_{\text{even } k} f(k) = \sum_{k=0}^n \frac{1^k + (-1)^k}{2} f(k)$

Power Series $\frac{1}{(1-x)^r} = \sum_{n=0}^{\infty} \binom{n+r-1}{r-1} x^n = \sum_{n=0}^{\infty} H_n^r x^n,$

$1+x+x^2+\dots+x^n = \frac{1-x^{n+1}}{1-x}, 1+x^n = \frac{1-x^{2n}}{1-x^n}$

Generalised Binomial Theorem $\binom{n}{r} = \frac{n(n-1)\dots(n-r+1)}{r!}.$

Vandermonde's Identity

$\sum_{i=0}^r \binom{m}{i} \binom{n}{r-i} = \binom{m}{0} \binom{n}{r} + \binom{m}{1} \binom{n}{r-1} + \dots + \binom{m}{r} \binom{n}{0} = \binom{m+n}{r}$
 $(1+x)^{m+n} = (1+x)^m (1+x)^n.$

Derangements $!n = n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! \dots$

Graph Theory

Graph Collection of vertices and edges; Finite, undirected, simple, with no loops. Denoted by $G = (V, E)$ where $V(G)$ is the vertex set and $E(G)$ is the edge set. $uv \in E(G), u \sim v$ is an edge. $v(G) = |V(G)|, e(G) = |E(G)|.$

Identical and Isomorphic Graphs G, H are identical if $V(G) = V(H), E(G) = E(H).$ $G \cong H$ if there is a bijection $f: V(G) \rightarrow V(H), u \sim v \in G \Leftrightarrow f(u) \sim f(v) \in H.$ This is an equivalence relation.

Complement $\bar{G}$ is the complement of G with the same vertex set $V(G)$ and edge set $\binom{V(G)}{2} \setminus E(G).$

Complete Graph K_n is the complete graph with n vertices and $\binom{n}{2}$ edges.

Petersen Graph Star connected to pentagon. 10 vertices, 15 edges, $deg(i) = 3.$

Hypercube Graph Q^d , vertices are $\{0, 1\}^d$, and edges are a bit flip, has 2^d vertices, $d \cdot 2^{d-1}$ edges.

Subgraph H is a subgraph of G if $V(H) \subset V(G), E(H) \subset E(G),$

$E(H) \subset \binom{V(H)}{2}.$ A spanning subgraph has $V(H) = V(G) (2^{\binom{n}{2}} \text{ of them}).$ An induced subgraph $G[S]$ on $S \subset V(G)$ has vertex set S and all edges from G

with both endpoints in $S (2^n \text{ of them}).$ There are $\sum_{k=0}^n \binom{n}{k} 2^{\binom{k}{2}}$ subgraphs.

Adjacency Matrix $n \times n$ symmetric matrix where $A_{i,j} = 1$ if $v_i v_j \in E(G), 0$ otherwise. Diagonal entries are zero.

Incidence Matrix v is incident to e if $v \in e. n \times m$ matrix where $M_{i,j} = 1$ if v_i is incident to $e_j, 0$ otherwise. Columns represent edges.

Relation between Representations $M_G M_G^T = \text{diag}(d(v_1), \dots, d(v_n)) + A_G.$

Number of Graphs $\binom{\binom{n}{2}}{m}$ labeled graphs. Asymptotically $/n!$ for unlabeled.

Neighbourhood $N(v)$ is the set of all u adjacent to $v.$

Degree $d(v) = |N(v)|.$ Degree sequence: List of all vertex degrees, often sorted in non-increasing order. $\delta(G)$: smallest degree, $\Delta(G)$: largest degree, k -regular: all vertices have degree $k.$ $\bar{d}(G) = \frac{1}{|V(G)|} \sum_{v \in V(G)} d(v)$: average degree.

Handshake Lemma $\sum_{v \in V(G)} d(v) = 2e(G).$ $\bar{d}(G) = \frac{2e(G)}{v(G)}.$ The number of odd degree vertices is even.

Bipartite Graph A graph is bipartite if $V = A \cup B, A \cap B = \emptyset,$ there are no edges within A or $B.$ A k -regular bipartite graph has $|A| = |B|.$

Graphic Sequences Degree sequence of some graph.

Havel-Hakimi Theorem The non-increasing sequence $(d_1, \dots, d_n)$ is graphic iff $(d_2 - 1, d_3 - 1, \dots, d_{d_1+1} - 1, d_{d_1+2}, \dots, d_n)$ is graphic. Repeatedly remove largest degree d_1 and subtract 1 from the next d_1 degrees, until obvious.

Erdős-Gallai Theorem A non-increasing sequence is graphic iff it has an even sum and for all $k \in \{1, \dots, n\}, \sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min\{d_i, k\}.$

Cliques $\omega(G)$ is the number of vertices in the largest clique of $G.$ $\alpha(G)$ is the size of the largest independent set in $G.$ $\alpha(G) = \omega(\bar{G}), \omega(G) = \alpha(\bar{G}).$

Graph Families C_n : cycle, P_n : path, $K_{m,n}$: complete bipartite.

Kneser Graph $KG(n, k)$ is the graph where vertices are all k -element subsets of n and edges are between disjoint k -subsets.

Erdős-Ko-Rado Theorem For $n \geq 2k, \alpha(KG(n, k)) = \binom{n-1}{k-1}.$

Graph Colourings A proper k -Colouring is $c: V(G) \rightarrow \{1, \dots, k\}$ such that $uv \in E(G) \Rightarrow c(u) \neq c(v).$ The chromatic number $\chi(G)$ is the smallest k such that G has a proper k -colouring.

Lovász's Theorem $\chi(KG(n, k)) = n - 2k + 2.$

Bounding Chromatic Number For every $G, \chi(G) \geq \omega(G), \chi(G) \geq \frac{|V(G)|}{\alpha(G)}.$

For m edges, $\chi(G) \leq \frac{1}{2} + \sqrt{2m + \frac{1}{4}}. \chi(G) \leq \Delta(G) + 1.$

Walks For length $k,$ non-empty alternating sequence $v_0, e_0, \dots, e_{k-1}, v_k$ with $e_i = \{v_i, v_{i+1}\} \in E(G).$ If $v_0 = v_k,$ the walk is closed.

Connectedness All pairs have a walk between them, equivalence relation.

Connected Component Partition of $V(G)$ based on connectedness.

Counting Walks $(A^k)_{ij}$ gives the number of walks of length k from i to $j.$

Connected Components Let D be the diagonal matrix of degrees. The Laplacian matrix of G is $L = D - A.$ Then the number of connected components of G is $nullity(L)$ (number of free variables in RREF).

Paths and Trails A trail is a walk with no repeated edges. If a walk or trail has no repeated vertices, it is a path. Every $u - v$ walk contains a $u - v$ path.

Distance $d_G(x, y)$ is the length of a shortest path connecting x and y, ∞ if it does not exist.

Diameter $diam(G) = \max_{x, y \in V(G)} d(x, y).$

Breadth First Search Expand layer by layer by adding each set of undiscovered neighbours to the search frontier.

Vertex Cut $S \subset V(G), G - S$ is disconnected. If $|S| = 1,$ it is a cut vertex.

Vertex Connectivity $\kappa(G) = \min\{|S| : S \subset V(G), G - S \text{ is disconnected}\}.$ k -connected: $\kappa(G) \geq k$ or deleting $k - 1$ vertices leaves it connected.

Girth / Circumference Minimum / Maximum length of cycle contained in $G.$

Acyclic Graph Graph with no cycle as a subgraph.

Cycle Proposition Every 2-connected graph contains a cycle.

r-Partite G is r -partite if it can be partitioned into $V(G) = V_1 \cup \dots \cup V_r.$

Bipartite Proposition A graph is bipartite iff it contains no odd cycles.

Eulerian Circuit Closed trail using every edge exactly once.

Euler's Theorem A graph is Eulerian iff all vertices have even degrees.

Depth First Search Recursively visit every unvisited neighbour before backtracking. Find cycles by checking for back edges to already visited nodes.

Hamiltonian A Hamiltonian cycle is a closed walk that visits every vertex exactly once. Path is similar. Q^n is Hamiltonian, while the Petersen graph is not.

Dirac's Theorem If G has $n \geq 3$ and $\delta(G) \geq n/2,$ then G is Hamiltonian. For even $n,$ if $\delta(G) \geq n/2,$ then G has a perfect matching.

Ore's Theorem If G has $n \geq 3$ and every non-adjacent pair u, v satisfies

$deg(u) + deg(v) \geq n,$ then G is Hamiltonian.

Closure Add edges where non-adjacent u, v have $deg(u) + deg(v) \geq n.$

Bondy-Chvátal Theorem G is Hamiltonian iff its closure $Cl(G)$ is Hamiltonian.

Forest / Tree Acyclic Graph. A tree is a connected forest.

Leaf Vertex with degree 1. Removal does not disconnect the graph.

Characterisations of Trees TFAE

- T is a tree.
- Any two vertices of T are connected by a unique path in $T.$
- T is minimally edge connected (removal of any edge disconnects the tree).

4. T is maximally acyclic (acyclic but any new edge creates a cycle).

5. T is a connected graph on n with $n - 1$ edges.

Vertex Ordering The vertices of a tree T can be ordered such that the i th vertex v_i has a unique neighbour in $\{v_1, \dots, v_{i-1}\}.$ Every tree has at least one leaf. A connected graph on n vertices is a tree iff it has $n - 1$ edges.

Embedding Trees If T is a tree on k vertices, and G is a graph with $\delta(G) \geq k - 1,$ then G contains a subgraph isomorphic to $T.$

Erdős-Sós Conjecture Every graph G on $n \geq k$ vertices having average degree $\bar{d}(G) > k - 2$ must contain every tree of k vertices.

Rooted Tree Tree with a fixed vertex called the root.

Spanning Tree Subgraph that is a tree and includes all vertices of $G.$ Every connected graph G has a spanning tree.

BFS/DFS Tree Spanning tree obtained by exploring a graph level by level from a root vertex / exploring as far as possible along each branch before backtracking.

Minimum Spanning Tree Spanning tree T minimising $w(T) = \sum_{e \in E(T)} w(e).$

Safe Edge Let A be a set of edges included in a MST that is being built. An edge is safe if $A \cup \{(u, v)\}$ is a subset of some final MST.

Cut Property Let A be a subset of some MST. For any cut $(S, V - S)$ that respects $A,$ if e is a minimum weight edge crossing the cut, then e is safe for $A.$

Kruskal's Algorithm Sort edges by non-decreasing weight. Then for each edge, if adding it does not create a cycle, add it to the MST.

Prim's Algorithm Start from an arbitrary vertex. Then repeatedly add the minimum weight edge that connects to new vertices.

Cycle Property For any cycle C the maximum weight edge e is not in any MST.

Reverse-Delete Algorithm Start with all edges, sorted in non-increasing order.

Then for each edge, if the graph without the edge is still connected, remove it.

Cayley's Theorem The number of distinct labeled trees on n vertices is $n^{n-2}.$

Prüfer Code

- Bijection between labeled trees on n vertices and sequences of length $n - 2$ with entries from $\{1, \dots, n\}.$
- Encoding: Repeatedly find the leaf with the smallest label, add its neighbour to the set, then remove the leaf.
- Decoding: Repeatedly find smallest element not in the sequence $P,$ then add edge between it and the first element of $P,$ then remove both.

Degree Sequence Formula The number of labeled trees on n vertices is $\frac{(n-2)!}{(d_1-1)! \dots (d_n-1)!}$ by multinomial coefficients.

Cofactor The (i, j) -cofactor of a matrix is $(-1)^{i+j}$ times the determinant of the submatrix obtained by deleting row i and column $j.$

Kirchhoff's Matrix Tree Theorem The number of spanning trees equals any cofactor of the Laplacian matrix $L(G).$ $\tau(G) = (-1)^{i+j} \det(L_{ij})$ where L_{ij} is L with row i and column j deleted.

Oriented Incidence Matrix $n \times m$ matrix where each column corresponds to an edge and for edge $(i, j),$ put $+1$ in row $i, -1$ in row $j,$ and 0 elsewhere.

Incidence Matrix and Laplacian $L = DD^T$

Binet-Cauchy Theorem If $A, B,$ are $n \times m$ with $m \leq n,$ then $\det(AB) = \sum_{S \subseteq [n], |S|=m} \det(A_{[m], S}) \det(B_S, [m])$ where the sum is over all m -element subsets of columns.

Kirchhoff Lemma For the incidence matrix with one row deleted; for any subset S of $n - 1$ edges, $\det(D_S) = \pm 1$ if S forms a spanning tree or 0 otherwise.

Weighted Matrix Tree Theorem For a weighted graph, the sum of weights of all spanning trees where tree weight = product of edge weights equals any cofactor of the weighted Laplacian $L_{ij} = \sum_{k \neq i} w_{ik}$ if $i = j, -w_{ij}$ otherwise.

This question is evil, but it's okay because I am evil.

students

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